

Experiment No. 7

Title

Mounting and Unmounting Devices in Linux

Aim

The problem is to identify storage devices in Linux, mount and unmount file systems manually, and understand the significance of mounting points and device management in system administration and cyber security environments.

Theory

Linux File System and Mounting

Linux follows a single-rooted directory structure where all files and devices are accessible through /. Storage devices must be mounted to a directory (mount point) before their contents can be accessed.

Device Files

Linux represents hardware devices as files located in the /dev directory. Common examples include:

- /dev/sda, /dev/sdb – Hard disks
- /dev/sda1, /dev/sdb1 – Disk partitions

Mount Points

A mount point is an empty directory where a device's file system is attached.

Example:

- /mnt
- /media

Unmounting Devices

Unmounting safely disconnects the device from the file system, ensuring all data is written properly and preventing corruption.

COMMANDS USED-

lsblk	List block devices
fdisk -l	Display disk partition information
mount	Mount a file system
umount	Unmount a file system
df -h	Display mounted file systems
mkdir	Create mount point

Procedure / Steps

A. Identifying Devices Open the terminal.

List available block devices:

```
lsblk
```

Display detailed disk information:

```
sudo fdisk -l
```

```
(amanyadav@kali-vm)-[~]
$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda          8:0    0   8G  0 disk
├─sda1       8:1    0  7.5G  0 part /
├─sda2       8:2    0    1K  0 part
└─sda5       8:5    0 480M  0 part [SWAP]
sr0         11:0    1 1024M  0 rom

(amanyadav@kali-vm)-[~]
$ sudo fdisk -l
[sudo] password for amanyadav:
Disk /dev/sda: 8 GiB, 8589934592 bytes, 16777216 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x59bfd4e6

Device      Boot      Start         End      Sectors  Size Id Type
/dev/sda1   *          2048     15790079   15788032    7.5G 83 Linux
/dev/sda2             15792126     16775167     983042    480M  f W95 Ext'd (LBA)
/dev/sda5             15792128     16775167     983040    480M 82 Linux swap / Solaris
```

B. Creating a Mount Point

Create a directory to be used as a mount point:

```
sudo mkdir /mnt/us
```

C. Mounting a Device

Mount a partition (example /dev/sdb1) to the mount point:

```
sudo mount /dev/sdb1 /mnt/us
```

Verify the mount:

```
df -h mount
```

```
(amanyadav@kali-vm)-[~]
$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            2.2G   0    2.2G   0% /dev
tmpfs           464M  996K  463M   1% /run
/dev/sda1       7.4G  6.5G  507M  93% /
tmpfs           2.3G  8.0K  2.3G   1% /dev/shm
none            1.0M   0    1.0M   0% /run/credentials/systemd-journald.service
tmpfs           2.3G  4.0K  2.3G   1% /tmp
tmpfs           464M  2.8M  461M   1% /run/user/1000

(amanyadav@kali-vm)-[~]
$ mount
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,relatime)
proc on /proc type proc (rw,nosuid,nodev,noexec,relatime)
udev on /dev type devtmpfs (rw,nosuid,relatime,size=2266840k,nr_inodes=566710,mode=755,inode64)
devpts on /dev/pts type devpts (rw,nosuid,noexec,relatime,gid=5,mode=600,ptmxmode=000)
tmpfs on /run type tmpfs (rw,nosuid,nodev,noexec,relatime,size=474416k,mode=755,inode64)
/dev/sda1 on / type ext4 (rw,relatime,errors=remount-ro)
securityfs on /sys/kernel/security type securityfs (rw,nosuid,nodev,noexec,relatime)
tmpfs on /dev/shm type tmpfs (rw,nosuid,nodev,inode64)
cgroup2 on /sys/fs/cgroup type cgroup2 (rw,nosuid,nodev,noexec,relatime,nsdelegate,memory_recursiveprot,memory_hugetlb_accounting)
none on /sys/fs/pstore type pstore (rw,nosuid,nodev,noexec,relatime)
bpf on /sys/fs/bpf type bpf (rw,nosuid,nodev,noexec,relatime,mode=700)
systemd-1 on /proc/sys/fs/binfmt_misc type autofs (rw,relatime,fd=41,pgrp=1,timeout=0,minproto=5,maxproto=5,direct,pipe_ino=5410)
tracefs on /sys/kernel/tracing type tracefs (rw,nosuid,nodev,noexec,relatime)
mqueue on /dev/mqueue type mqueue (rw,nosuid,nodev,noexec,relatime)
```

D. Accessing the Mounted Device

Navigate to the mounted directory:

```
cd /mnt/usb ls
```

E. Unmounting the Device

Exit the mounted directory:

```
cd ~
```

Unmount the device:

```
sudo umount /mnt/usb
```

Verify unmounting:

```
df -h
```

Outcomes

- ✓ Storage devices and partitions are listed successfully.
- ✓ Device is mounted to the specified mount point.
- ✓ Files inside the mounted device are accessible.
- ✓ Device is safely unmounted without errors.

Result

- ❖ Storage devices were successfully identified, mounted, accessed, and unmounted in Linux. The student gained practical understanding of Linux file system mounting and device management.